Liouville's Theorem and the Fundamental Theorem of Algebra

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of liouville's theorem and the fundamental theorem of algebra, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

CA-02 / Complex Analysis / Report v1

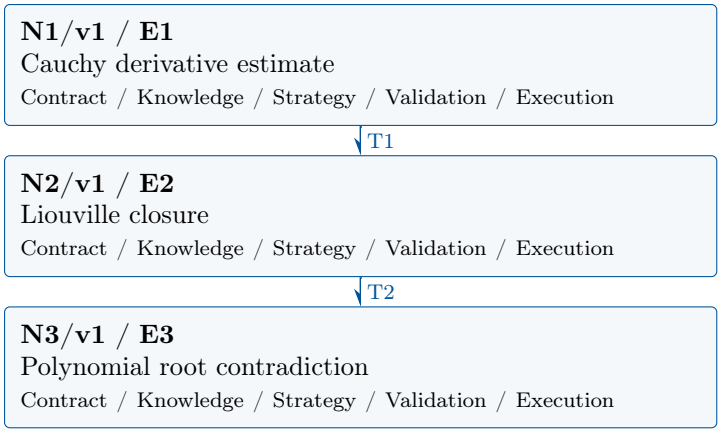
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Prove every bounded entire function is constant. Deduce that every nonconstant complex polynomial has a complex root.*

A. Foundations and permitted imports

Import CA-01 and differentiation under its contour integral on a disk: $f'(z_0) = (2\pi i)^{-1} \oint f(z)/(z - z_0)^2 \, dz$. The differentiation is justified by a locally uniform denominator bound away from the contour. Use connectedness of C and polynomial leading-term estimates; do not import root existence. Also use the real mean value theorem on a line segment and the extreme value theorem for continuous real functions on compact disks.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. Every derivative of order 1 vanishes.

Receiving activity. N2: Liouville closure

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Derivative zero everywhere; prior exports E1/v1

T2: N2/E2/v1 to N3. Every bounded entire function is constant.

Receiving activity. N3: Polynomial root contradiction

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Nonconstant p of degree d with leading coefficient $a_d \neq 0$; prior exports E2/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Cauchy derivative estimate

Object and role. Let a contour radius grow while the bound stays fixed.

Inputs. Entire f with $|f| \le M$ globally; foundational imports stated in §1

C — Contract

Own E1/v1: Every derivative of order 1 vanishes.

K — Knowledge

Retain Entire f with $|f| \le M$ globally; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Let a contour radius grow while the bound stays fixed.

V — Validation

A bound depending on R cannot be sent to infinity this way.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 For any z_0 and $R > 0$ the differentiated formula applies on the disk about z_0.

N1.2 Its length $2\pi R$ and denominator modulusR ^{2} give $|f'(z_0)| \le M/R$.

N1.3 The same M holds for every radius since f is globally bounded.

N1.4 Let $R \to \infty$; $f'(z_0) = 0$ for every z_0.

Export E1/v1. Every derivative of order 1 vanishes.

Acceptance gate. A bound depending on R cannot be sent to infinity this way. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: A bound depending on R cannot be sent to infinity this way. Recheck dependent nodes N2, N3.

B. N2/v1: Liouville closure

Object and role. Use the connected domain to obtain one constant.

Inputs. Derivative zero everywhere; prior exports E1/v1

C — Contract

Own E2/v1: Every bounded entire function is constant.

K — Knowledge

Retain Derivative zero everywhere; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Use the connected domain to obtain one constant.

V — Validation

Zero derivative on a disconnected domain only gives componentwise constancy.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 For any z,w, consider $t \mapsto f(z + t(w - z))$ on[0,1].

N2.2 Its real parameter derivative is $f'(z + t(w - z))(w - z) = 0$; real and imaginary parts are constant by the real mean value theorem.

N2.3 Thus $f(w) = f(z)$.

N2.4 This proves Liouville's theorem on C.

Export E2/v1. Every bounded entire function is constant.

Acceptance gate. Zero derivative on a disconnected domain only gives componentwise constancy. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Zero derivative on a disconnected domain only gives componentwise constancy. Recheck dependent nodes N3.

C. N3/v1: Polynomial root contradiction

Object and role. Apply Liouville to a hypothetical entire reciprocal.

Inputs. Nonconstant p of degree d with leading coefficient $a_d \neq 0$; prior exports E2/v1

C — Contract

Own E3/v1: Every nonconstant complex polynomial has a complex root.

K — Knowledge

Retain Nonconstant p of degree d with leading coefficient $a_d \neq 0$; prior exports E2/v1; keep their assumptions and versions attached.

S — Strategy

Apply Liouville to a hypothetical entire reciprocal.

V — Validation

Bound the reciprocal on the compact interior as well as at infinity.

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Assume p has no zero.

N3.2 Then 1/p is entire.

N3.3 Choose R so for $|z| \geq R$ the sum of lower-degree terms is at most $|a_d||z|^d/2$.

N3.4 Thus $|p(z)| \geq |a_d||z|^d/2$ and $|1/p(z)| \leq 2/(|a_d|R^d)$ outside.

N3.5 On the closed radius-R disk continuity and nonvanishing give a positive minimum of|p|, hence a finite reciprocal bound.

N3.6 Liouville makes 1/p constant, whereas it tends to 0 at infinity and is nowhere zero, contradiction.

N3.7 Therefore p has a root.

Export E3/v1. Every nonconstant complex polynomial has a complex root.

Acceptance gate. Bound the reciprocal on the compact interior as well as at infinity. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Bound the reciprocal on the compact interior as well as at infinity. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Cauchy derivative estimate

1. For any z_0 and $R > 0$ the differentiated formula applies on the disk about z_0.
2. Its length $2\pi R$ and denominator modulusR ^{2} give $|f'(z_0)| \leq M/R$.
3. The same M holds for every radius since f is globally bounded.
4. Let $R \to \infty$; $f'(z_0) = 0$ for every z_0.

B. §4.N2 — Liouville closure

1. For any z,w, consider $t \mapsto f(z + t(w - z))$ on[0,1].
2. Its real parameter derivative is $f'(z + t(w - z))(w - z) = 0$; real and imaginary parts are constant by the real mean value theorem.
3. Thus $f(w) = f(z)$.
4. This proves Liouville's theorem on C.

C. §4.N3 — Polynomial root contradiction

1. Assume p has no zero.
2. Then 1/p is entire.
3. Choose R so for $|z| \geq R$ the sum of lower-degree terms is at most $|a_d||z|^d/2$.
4. Thus $|p(z)| \geq |a_d||z|^d/2$ and $|1/p(z)| \leq 2/(|a_d|R^d)$ outside.
5. On the closed radius-R disk continuity and nonvanishing give a positive minimum of|p|, hence a finite reciprocal bound.
6. Liouville makes 1/p constant, whereas it tends to 0 at infinity and is nowhere zero, contradiction.
7. Therefore p has a root.

□

V. WORKED ORCHESTRATION

The entire function e^z is nonconstant and is unbounded along positive real z, consistent with Liouville. For $p = z^2+1$, the root-existence conclusion holds and the explicit $factors(z-i)(z+i)$ illustrate it.

A. Boundary and hypothesis audit

The function1/z is bounded $on\{|z| \geq 1\}$ but has a pole at 0, so it is not a counterexample to Liouville. Validation identifies the missing entire-domain condition; a bounded function on the unit disk can also be nonconstant.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

The function1/z is bounded $on\{|z| \geq 1\}$ but has a pole at 0, so it is not a counterexample to Liouville. Validation identifies the missing entire-domain condition; a bounded function on the unit disk can also be nonconstant.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- Harvard complex analysis: Cauchy formula, argument principle, and Riemann mapping lectures.